FlowBoard

1. Problem Statement

To empower our engineers to think deeply about **state management, design trade-offs, and AI-native development practices**, we are launching a focused assignment to build a **lightweight Kanban board application**, codenamed *FlowBoard*.

The vision is to create a simple task management tool that mimics the core interactions of Trello but with reduced scope, focusing on **component design, drag-drop interactions, and iterative AI-assisted development**.

Please note: For the assignment, please use localStorage as the persistence layer, no API integration is required.

2. Core Functional Requirements

- **Columns**: The board must have **three fixed columns** – *To Do*, *In Progress*, *Done*.
- **Add Task**: Users must be able to add a task (title only) to the *To Do* column.
- **Move Task**: Tasks can be moved across columns (via drag-drop OR buttons).
- **Delete Task**: Users can delete a task from any column.
- **Task State**: Tasks must be stored in component state.

3. UI Layout

The application should follow a **three-column layout**:

- **Left (33%)**: To Do column – add new tasks here.
- **Middle (33%)**: In Progress column.
- **Right (33%)**: Done column.

Each column displays its tasks vertically. Tasks must be interactive (move/delete).

4. Technical Constraints & Rules

- No external drag-drop libraries (`react-dnd`, `interact.js`, `konva.js`) allowed.
- Must use React/Angular/Vue/React Native core features only.
- Code must compile and run locally using `npm run start`.
- App must not depend on any external API.

5. Deliverables

Create a private GitHub repo and share with the below members:

- dmistryTal
- nileshmallick1606
- Sachin-Salunke-Talentica

Your submission will be a **single private GitHub repository** containing:

- **Source Code**: Complete, runnable application.
- **README.md**: Clear build/run instructions.
- **PROJECT_STRUCTURE.md**: Explanation of folder/module layout.
- **ARCHITECTURE.md**:
  - Chosen architectural pattern.
  - Component hierarchy diagram.
  - State management explanation.
  - Why drag-drop was implemented the chosen way.
- **CHAT_HISTORY.md**: Prompt logs with coding assistant, including iterations and trade-offs.
- **Video (5–7 min)**: Explaining design, architecture, AI journey, key decisions, and demo.
  - **Upload the video on your one drive and share the access with the below members:**
    - Dipen.mistry@talentica.com
    - Nilesh.Mallick@talentica.com
    - Sachin.Salunke@talentica.com